

Name:

Roll No.

Section

1. Find the output of the Matlab code

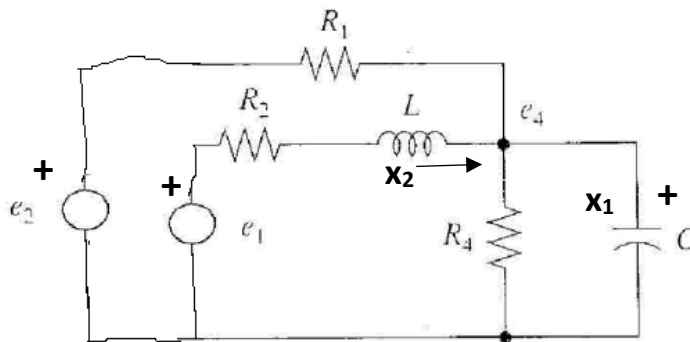
```
b = [2 0 0 1 0];  
a = [1 0 1];  
[p,r,k] = residue(b,a)
```

$$p = 0.5 \pm j1$$

$$r = \pm j1$$

$$k = [2 \quad 0 \quad -2]$$

2. Find the A,B C and D matrices of the following network $R_1 = R_4 = 2 \Omega$, $R_2 = 1 \Omega$, $L = 1 \text{ H}$, $C = 1 \text{ F}$
The output are the voltages across R_1 , R_2 and the current through C



$$D = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0.5 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0.5 \\ 1 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$$